18 Methodologies for in Vitro Cultivation of Arbuscular Mycorrhizal Fungi with Root Organs

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1 Introduction

The monoxenic culture of arbuscular mycorrhizal (AM) fungi has markedly improved our understanding of the symbiosis. In the past 15 years, increasing amounts of literature have been devoted to this intimate plant–fungal association using various AM fungi in vitro cultivation systems, with different hosts, AM fungal propagules and growth media. The proportion of papers published using either in vitro, axenic, monoxenic, root organ culture or ROC as keywords, relative to the overall literature dealing with AM fungi (ISI web of Science http://www.isinet.com)[4], was less than 1% in the years 1987–1989, increasing to approximately 5% in the subsequent 6 years (1990–1995), to reach a plateau at 8% from 1996 to present. The invariable proportion between papers using AM fungi in vitro systems and complete literature on AM fungi since 1996 until today suggests that the use of this system still remains in the hands of a limited number of researchers, and that new progress is necessary to reach a broader audience. If one agrees that the axenic culture of AM fungi remains the major challenge at the start of this new millennium, the present diffusion of clear protocols on AM fungi in vitro culture techniques may render this technology more widely accessible and secure its broad and reliable dissemination. Indeed, as stated by Bago and Cano (see Chap. 7) … "*AM (fungi) monoxenics are far*

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[4]Keywords used for the overall literature on endomycorrhizal fungi: 'AM fungi' OR 'AM symbiosis' OR 'VAM fungi' OR 'VAM symbiosis' OR 'endomycorrhiz*' OR 'vesicular-arbusc*' OR 'vesicular arbuscular' OR 'arbuscular' OR (*Glomus* AND mycorrhiz*). Keywords used for literature related to the endomycorrhizal fungi monoxenically cultured: ['AM fungi' OR 'AM symbiosis' OR 'VAM fungi' OR 'VAM symbiosis' OR 'endomycorrhiz*' OR 'vesicular-arbusc*' OR 'vesicular arbuscular' OR arbuscular OR (*Glomus* AND mycorrhiz*)] AND ('in vitro' OR 'monoxenic' OR 'axenic' OR 'root organ culture' OR 'transformed roots')

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more than just a routine technique ... strict protocols should be followed to successfully achieve it and ... some training/expertise on AM establishment, fungal colony development and hyphal morphogenesis under such conditions is mandatory for researchers aiming to use this technique, to be able to certify the quality of the material obtained and, consequently, the reliability and accurateness of the results obtained." This chapter is not aimed to propose a literature overview but rather to provide detailed protocols to succeed in the monoxenic culture of AM fungi. It is obvious that for every technique different protocols (disinfection process, growth medium, choice of propagule, etc.) have been published, and we cannot discuss all of these in detail. Therefore, we choose to describe the routine techniques used in GINCO (*G*lomeromycota IN Vitro COllection) with, at each step, references to other papers using other approaches, so giving the reader the broadest perception of the techniques that presently exist and which are the most appropriate for their research interest.

2 Process Description

The process of obtaining and maintaining monoxenic cultures of AM fungi can be separated into four main steps. These are the selection of the adequate AM fungal propagules (see Sect. 6.1), the sampling, disinfection and incubation of the propagules on a suitable growth medium (see Sect. 6.2), the association of the propagules with a suitable host root (see Sect. 7), and the subcultivation of the AM fungi (see Sect. 8). Prior to these four steps are the selection of the appropriate culture system (see Sect. 3), the preparation of the synthetic culture media (see Sect. 4), and the management of the host root, i.e. transformation and subcultivation (see Sect. 5; Fig. 1).

3 Selection of the Culture System

Basically, two culture systems are used: the mono-compartmental system in square or round Petri plates, and the bi-compartmental system in round Petri plates.

The first system consists of a mono-compartmental Petri plate filled with a growth medium (see Sect. 4), on which is placed a contaminant-free, actively growing excised root together with AM fungal propagules. This system was developed in the mid-1970s (Mosse and Hepper 1975) and since then has been applied with success to numerous *Glomus* species (Table 1). Recent reports have also demonstrated its applications to *Scutellospora*

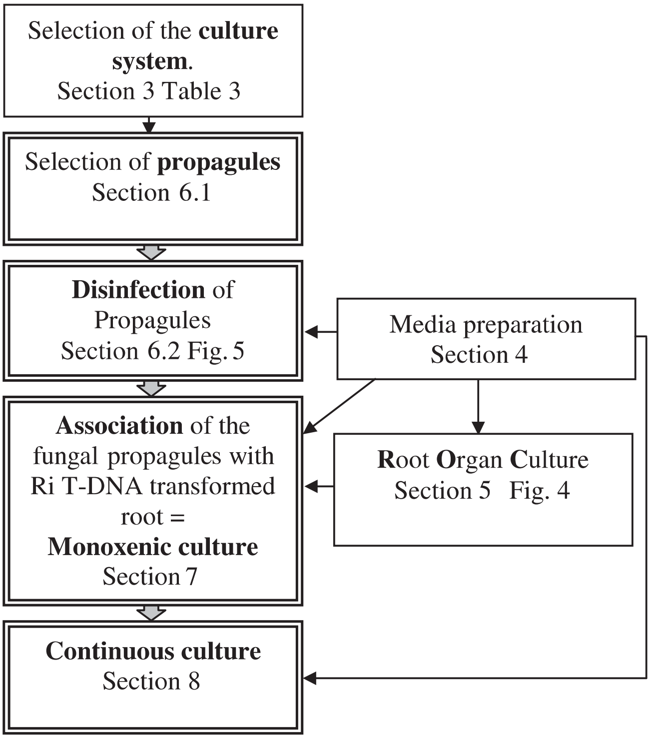


Fig. 1. General scheme of the monoxenic culture process

reticulata (de Souza and Declerck 2003) and *Acaulospora rehmii* (Dalpé and Declerck 2002).

This system was slightly modified for the cultivation of *Gigaspora* species (*Gi. margarita*, *Gi. roseae*, *Gi. gigantea*; Table 1) by placing a Ri T-DNA transformed carrot root – having negative geotropism – in the upper part of a square Petri plate set vertically, with the germinated spore just below (Fig. 7F; Bécard and Fortin 1988; Bécard and Piché 1992; Diop et al. 1992). Since the germ tube growth of these *Gigaspora* species is also negatively geotropic, the germinating hyphae are oriented towards the root, thus facilitating contact and colonization (see Sect. 7). The root growth is restricted to the upper part of the plate while after colonization the mycelium develops in the lower part. It was observed that the sporulation mainly occurs in the section with less root development.

The second system (also named split system) consists of a bi-compartmental Petri plate, with a proximal compartment in which the mycorrhizal root develops and containing a synthetic growth medium (see Sect. 4), and a distal compartment in which only the mycelium is allowed to grow

Table 1. General overview of the AM fungi cultured in the monoxenic system with the different options available for the medium, culture system, propagule type, etc. The mentioned citations refer to the first and most descriptive system with the different options mentioned (*continued on next page*)

Species	Starter propagules	Host root	Medium	Culturing model	Production of mature spores (size, μm)[a]	Continuous culture[b]	Citations
AMF species not producing vesicles							
Gigaspora gigantea (Nicolson & Gerd.) Gerd. & Trappe	Spores	T. *Daucus carota* L.[1]	M	Mono (incubated with angle)	Yes	No	Gadkar et al. (1997)
	Spores	T. *Daucus carota* L.	G+T (see citation)	Mono	Yes	No	Mosse (1988)
Gigaspora margarita Becker & Hall	Spores	NT. *Lycopersicon esculentum* Mill.[2]	MS	Mono	Yes	No	Miller-Wideman and Watrud (1984)
	Spores	T. *Daucus carota* L.	M	Mono (incubated with angle)	Yes	No	Gadkar et al. (1997)
	Spores	T. *Daucus carota* L.	M	Mono	Yes	No	Karandashov et al. (1999)
		NT. *Linum usitatissimim* L.[3] NT. *Tagetes pastula* L.[4] NT *Althaea officinalis* L.[5]					
Gigaspora roseae Nicolson & Schenck	Spores	T. *Daucus carota* L.	M	Mono (incubated vertically; square Petri plates)	Yes (mean= 300 μm)	No	Bécard and Fortin (1988), Bécard and Piché (1992), Diop et al. (1992)
Scutellospora reticulata (Koske, Miller, Walker) Walker & Sanders	Spores	T. *Daucus carota* L.	MSR	Mono	Yes (280–500, mean=376)	Yes	de Souza and Declerck (2003)
AMF species producing vesicles							
Acaulospora rehmii (Sieverding & Toro)	Spores	T. *Daucus carota* L.	MSR	Mono	Yes (112–162)	No	Dalpé and Declerck (2002)
Glomus caledonium (Nicoson & Gerd.) Trappe & Gerd.	Spores	T. *Daucus carota* L.	M (modified)	Mono	Yes (180–300)	Yes	Karandashov et al. (2000)
Glomus cerebriforme McGee	MRF	T. *Daucus carota* L.	M	Bi	Yes	Not mentioned	Samson et al. (2000)

Glomus clarum Nicolson & Schenck	Spores	T. *Daucus carota* L.	M	Mono	Yes (100–180, mean=130)	Yes	de Souza and Berbara (1999)
		T. *Trifolium repens* L.[6]					
Glomus constrictum Trappe	Spores	Plantlets of *Ziziphus mauritania*[7]	MS modified	Other	Not mentioned	No	Mathur and Vyas (1999)
Glomus deserticola Trappe, Bloss & Meng	MRF+spores	Plantlets of *Ziziphus mauritania*	MS modified	Other	Not mentioned		Mathur and Vyas (1995)
Glomus etunicatum Becker & Gerd	Spores	T. *Daucus carota* L.	M	Mono	Yes (40–130)	Yes	Pawlowska et al. (1999)
Glomus fasciculatum (Thaxter sensu Gerd.) Gerd. & Trappe emend. Walker & Koske	MRF	NT. *Fragaria x Ananassa* Duchesne.[8]	SR	Mono	Yes	Yes	Strullu and Romand (1986)
		NT. *Alium cepa* L.[9] NT. *Solanum lycopersicum* Mill[10]					
	MRF	T. *Daucus carota* L.	MSR	Mono	(60–90)	Yes	Declerck et al. (1998)
Glomus fistulosum Skou & Jakobson	Spores	T. *Fragaria x Ananassa* Duchesne	M	Mono	No	No	Nuutila et al. (1995)
Glomus intraradices Schenck & Smith	Isolated vesicles from MRF	NT. *Solanum lycopersicum* L.	SR	Mono	Yes	Yes	Strullu and Romand (1987)
	Spores	T. *Daucus carota* L.	M	Mono	Yes (85)	Yes	Chabot et al. (1992)
		NT. *Solanum lycopersicum* L.					
	MRF	NT. *Solanum lycopersicum* L.	M	Mono	Yes (65–80, mean=80)	Yes	Diop et al. (1994a, 1994b)
	Spores	T. *Daucus carota* L.	M	Bi	Yes	Yes	St-Arnaud et al. (1996)
	Spores+MRF	T. *Daucus carota* L.	M	Mono	Yes	Not mentioned	Karandashov et al. (1999)
	MRF	T. *Daucus carota* L.	MSR	Mono	Yes (80–100)	Yes	Declerck et al. (1998)
	Spores	T. *Medicago truncatula* Gaertn.[11]	M	Mono	Yes	Not mentioned	Boisson-Denier et al. (2001)
	MRF	T. *Daucus carota* L.	M	Bi	Yes	Yes[c]	Douds (2002)

Table 1. (*continued*)

Species	Starter propagules	Host root	Medium	Culturing model	Production of mature spores (size, µm)[a]	Continuous culture[b]	Citations
Glomus mosseae Nicolson & Gerdemann	Spores isolated from sporocarps	NT. *Lycopersicum esculentum* Mill. NT. *Trifolium pratense* L[12]	Modified White's medium	Mono	No	No	Mosse and Hepper (1975)
	Spores	T. *Daucus carota* L. T. *Convolvulus sepium* L.[13] T. *Solanum tuberosum* L.[14] T. *Vigna unguiculata* Walp.[15]	(Mugnier and Mosse 1987)	Bi (modified; see citation)	No	No	Mugnier and Mosse (1987)
	Spores+MRF	T. *Daucus carota* L.	M	Mono	No	No	Karandashov et al. (1999)
	Spores	T. *Daucus carota* L.	M	Mono	No	No	Douds (1997)
	Spores	*Saccharum officinarum*[16] *Sorghum vulgare*[17]	M	Mono	No	No	Raman et al. (2001)
Glomus macrocarpum Tulasne & Tulasne	MRF	T. *Daucus carota* L.	MSR	Mono	Yes (160–180)	Yes	Declerck et al. (1998)
Glomus proliferum Dalpé & Declerck	MRF	T. *Daucus carota* L.	MSR	Mono	Yes (22, 40–66)	Yes	Declerck et al. (2000)
				Bi	(46)		
Glomus versiforme (Karsten) Berch	MRF	NT *Solanum lycipersicum* L.	M	Mono	Yes (mean=65)	Yes	Diop et al. (1994a)
	MRF	T. *Daucus carota* L.	MSR	Mono	Yes (60–80)	Yes	Declerck et al. (1998)

MRF = Mycorrhizal root fragment. M = Minimal medium (Bécard and Fortin, 1988). MSR = Modified Strullu-Romand medium (Declerck et al. 1996). SR = Strullu-Romand medium (Strullu and Romand 1986). G+T = Modification of the Goford and Torrey medium (Mosse 1988). Mono = Mono-compartmental system. Bi = Bi-compartmental system. Other = Modified *in vitro* culture system in flask with plantlets as host. T = Ri T-DNA transformed root with *A. rhizogenes*. NT = Non-transformed excised root.

1 = Carrot, 2 = Tomato, 3 = Flax, 4 = Marigold, 5 = Marshmallow, 6 = Clover, 7 = Indian jujube, 8 = Strawberry, 9 = Onion, 10 = Tomato, 11 = Barrelclover, 12 = Red clover, 13 = Bindweed, 14 = Potato, 15 = Cowpea, 16 = Sugar cane, 17 = Sorghum.

[a] If mentioned in the publication; [b] Yes (No) = With (Without) production of mature spores after life cycle establishment; [c] = Which consist by extracting the distal compartment and refill it with fresh medium

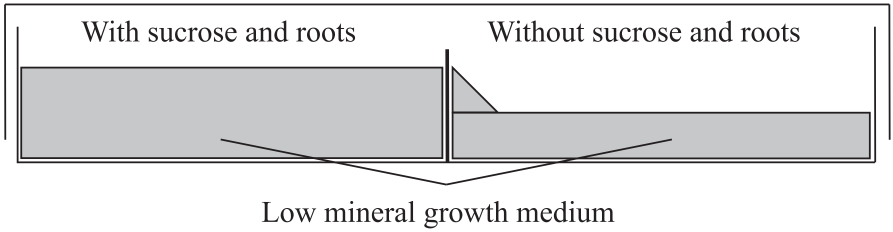


Fig. 2. Diagram of the bi-compartmental system from St-Arnaud et al. (1996). The mycorrhizal root grows in the proximal compartment and only the fungus is allowed to cross the separation wall and to develop in the distal compartment

on a similar synthetic medium, but lacking C source. Both compartments are physically separated by a plastic wall. Roots crossing the partition are trimmed at regular intervals. This system was developed by St-Arnaud et al. (1996; Fig. 2).

In the bi-compartmental system, the spore and mycelium density produced in the distal compartment is markedly higher in comparison to the proximal compartment, which is probably related to the absence of the root and the difference in availability of C (Fortin et al. 2002), making this system more productive than the mono-compartmental system. It is particularly adapted to the culture of *Glomus* species (Table 1). Recently, Douds (2002) has increased the spore production of a *G. intraradices* strain by repeated harvesting and gel replacement of the distal compartment.

Practical recommendations before starting a monoxenic culture:
Most of the steps described below are conducted under a laminar flow, under sterile conditions. A biohazard flow hood can also be suitable, but specific arrangements are necessary when a stereo microscope is used. The bench should be cleaned with ethanol. Similarly, all the material (needles, scalpels, forceps, etc.) necessary for monoxenic cultivation should be sterile or sterilized by flaming or in a bead sterilizer. Other materials (glassware, membranes, etc.) must be autoclaved.

4
Culture Media Preparation

The most widely used mineral media for monoxenic cultivation of AM fungi are the Minimal (M) medium (Bécard and Fortin 1988) and the Modified Strullu-Romand (MSR) medium (Declerck et al. 1996, from Diop 1995, modified from Strullu and Romand 1986). Both media result from the em-

Table 2. Comparison of the composition of minimal (M) medium and modified Strullu-Romand (MSR) medium (Fortin et al. 2002)

	M medium	MSR medium
$N(NO_3^-)$, µM	3,200	3,800
$N(NH_4^+)$, µM	–	180
P (µM)	30	30
K (µM)	1,735	1,650
Ca (µM)	1,200	1,520
Mg (µM)	3,000	3,000
S (µM)	3,000	3,013
Cl (µM)	870	870
Na (µM)	20	20
Fe (µM)	20	20
Mn (µM)	30	11
Zn (µM)	9	1
B (µM)	24	30
I (µM)	4.5	–
Mo (µM)	0.01	0.22
Cu (µM)	0.96	0.96
Ca panthotenate (µM)	–	1.88
Biotin (µM)	–	0.004
Pyridoxine (µM)	0.49	4.38
Thiamine (µM)	0.3	2.96
Cyanocobalamine (µM)	–	0.29
Nicotinic acid (µM)	4	8.10
Glycine (mg/l)	3	–
Myo-inositol (mg/l)	50	–
Sucrose (g/l)	10	10
pH (before sterilization)	5.5	5.5
Gelling agent (g/l)	5	3

piric modification of media usually used for in vitro plant culture and are equally successful for a range of AM fungi. The composition of M and MSR media is listed in Table 2. Differences between both media are discussed in Fortin et al. (2002). Similarly, a complete comparison of the successive media developed prior to the M and MSR media is given in Bécard and Piché (1992).

4.1 Material

Equipment

- Laminar flow hood
- Autoclave
- Balance (0.001 g)
- Magnetic stirrer
- pH-meter
- Glass bottles (50, 100, 500, 1000 ml)

Laboratory material

- Erlenmeyer flask
- Volumetric flask (50, 100, 500, 1000 ml)
- Petri plates
- Spoon
- Glass bottles (50, 100, 500, 1000 ml)

4.2 Stock Solutions for MSR Medium

The MSR medium routinely used in GINCO is detailed below. For the M medium preparation, refer to Bécard and Fortin (1988).

1. Solution 1: macroelements
   To prepare 1 l of solution, dissolve 73.9 g $MgSO_4{\cdot}7H_2O$, 7.6 g KNO_3, 6.5 g KCl and 0.41 g KH_2PO_4 in 500 ml of demineralized water. Adjust the volume to 1 l and store at 4 °C.

2. Solution 2: calcium nitrate
   To prepare 1 l of solution, dissolve 35.9 g $Ca(NO_3)_2{\cdot}4H_2O$ in 500 ml of demineralized water. Adjust the volume to 1 l and store at 4 °C.

3. Solution 3: vitamins
   To prepare 500 ml of solution, dissolve 0.09 g calcium panthotenate, 0.0001 g biotin, 0.1 g nicotinic acid, 0.09 g pyridoxine, 0.1 g thiamine and 0.04 g cyanocobalamine in 200 ml of demineralized water. Adjust the volume to 500 ml. Dispense in 5-ml fractions and store at −20 °C.

4. Solution 4: NaFeEDTA
   To prepare 500 ml of solution, dissolve 0.16 g NaFeEDTA in 200 ml of demineralized water. Adjust the volume to 500 ml, preserve from light and store at 4 °C.

5. Solution 5: microelements
   To prepare 500 ml of solution,

   a) Dissolve 1.225 g of $MnSO_4 \cdot 4H_2O$ (or 0.93 g of $MnSO_4 \cdot H_2O$) in 50 ml demineralized water. Adjust the volume to 100 ml.
   b) Dissolve 0.14 g of $ZnSO_4 \cdot 7H_2O$ in 50 ml of demineralized water. Adjust the volume to 100 ml.
   c) Dissolve 0.925 g of H_3BO_3 in 50 ml of demineralized water. Adjust the volume to 100 ml.
   d) Dissolve 1.1 g of $CuSO_4 \cdot 5H_2O$ in 30 ml of demineralized water. Adjust the volume to 50 ml.
   e) Dissolve 0.12 g of $Na_2MoO_4 \cdot 2H_2O$ in 50 ml of demineralized water. Adjust the volume to 100 ml.
   f) Dissolve 1.7 g of $(NH_4)_6Mo_7O_{24} \cdot 4H_2O$ in 50 ml of demineralized water. Adjust the volume to 100 ml.
   g) Mix the 100 ml of the manganese sulphate (a), zinc sulphate (b) and boric acid (c) solutions.
   h) Add 5 ml of copper sulphate (d) solution, 1 ml of sodium molybdate (e) and 1 ml of ammonium molybdate solution (f). Adjust the volume to 500 ml and store at 4 °C.

4.3 Medium Preparation

1. To prepare 1 l of MSR medium, mix 10 ml of solution 1 (macroelements) with 10 ml of solution 2 (calcium nitrate), 5 ml of solution 3 (vitamins), 5 ml of solution 4 (NaFeEDTA), 1 ml of solution 5 (microelements) and 10 g of sucrose.
2. After complete dissolution of the sucrose, adjust the volume to 1 l with demineralized water.
3. Adjust the pH to 5.5 with NaOH 1 M (rectify with HCl 1 M if necessary).
4. Add 3 g/l Gelgro (ICN Biochemicals, Cleveland, OH) or 5 g/l Phytagel™ or 8 g/l agar.
5. Autoclave for 15 min at 121 °C under 1 bar pressure.
6. After autoclaving, maintain the medium at 60 °C in a water bath.

Table 3. Medium quantities required in the different monoxenic systems

	Proximal or root compartment	Slope	Distal-compartment	Used preferably for
Mono-compartmental Petri plates (9-cm diameter)	Fill with ~ 35 ml MSR	No	No	*Scutellospora*, *Glomus*
Mono-compartmental Petri plates (squared 12×12 cm)	Fill with ~ 100 ml MSR	No	No	*Gigaspora*
Bi-compartmental Petri plates (9-cm diameter)	Fill the proximal compartment with ~ 25 ml MSR until the top of the separation	Create a slope of 45 °C with ~ 5 ml MSR without sugar	Fill with ~12 ml of MSR without sugar	*Glomus*

7. Pour the medium in the Petri plates under laminar flow, as indicated in Table 3.

For bi-compartmental Petri plates, it is important to promote spreading of mycelium from the proximal to the distal compartment. Two procedures can be used. In the first procedure, the proximal compartment is filled with the synthetic growth medium to the top of the separation wall. In the distal compartment, a slope (~ 45°) is realized with the same medium but lacking C. After solidification, the distal compartment is half-filled horizontally with the same medium lacking C (Fig. 2). In the second procedure, both compartments are filled 2 mm above the partition wall (Rufyikiri et al. 2003). After solidification, the medium in the distal compartment is cut along the partition wall with a scalpel and removed by using a spatula. The distal compartment is further replaced with a new medium lacking C at the level of the partition wall.

5 Host Root

5.1 Choice of Host Root

Various plants have been used to establish monoxenic cultures with AM fungi (see Chap. 7, Table 1). The first host roots used in monoxenic cultures were *Lycopersicon esculentum* Mill. (tomato) and *Trifolium pratense* L. (red clover) associated with *Glomus mosseae* Nicolson & Gerd. (Mosse and Hepper 1975), and *Fragaria x ananassa* Duchesne. (strawberry) and *Alium cepa* L. (onion) associated with several *Glomus* species (Strullu and Romand 1986; Table 1).

A natural genetic transformation of roots with the soil bacterium *Agrobacterium rhizogenes* Conn. was achieved decades ago (Riker et al. 1930; Ark and Thompson 1961), but only applied in mycorrhizal research since the mid-1980s (Mugnier and Mosse 1987). Since the development of the carrot hairy root line (*Daucus carota* L.), established by Bécard and Fortin (1988), this root has become the most widespread host for monoxenic cultivation of AM fungi. Transformed roots of strawberry (Nuutila et al. 1995) and tomato (Simoneau et al. 1994; Khaliq and Bagyaraj 2000) were also developed with fewer studies published. Recently, transformed roots of *Medicago truncatula* Gaern. have been obtained and associated with AM fungi (Boisson-Denier et al. 2001). It is probable that this association will receive increasing attention in the coming years, thanks to its usefulness as a legume model plant in molecular biology (Cook 1999; May and Dixon 2004). Transformed roots have several advantages over non-transformed roots for monoxenic cultivation. Their hormonal balance is modified, allowing profuse proliferation on synthetic media. It is generally accepted that this modification induces the production of growth hormones in the roots, thereby eliminating the necessity of incorporating of plant hormones into the culture medium. Stability of the transformation over time is dependent on the host cultivar and bacterial strain combination (Labour et al. 2003), but hairy roots used for further experiments have always tested positive.

In association with AM fungi, Ri T-DNA transformed roots show greater AM intraradical colonization and sustain higher extraradical hyphal development than non-transformed roots, which is an advantage for fungal production (Fortin et al. 2002). However, the pattern of colonization, distribution of vesicles, mycorrhizal spread and sporulation mechanisms – all the important growth stages of the fungus – can vary within different cultivars of the same host (Tiwari and Adholeya 2003). Furthermore, transformation of the same host cultivar with a different bacterial strain results in a different mycorrhizal susceptibility (Labour et al. 2003). More research is therefore needed to get a better insight into the physiological host preference/specificity of these fungi.

5.2
Host Root Transformation

Agrobacterium rhizogenes Conn., a gram-negative soil bacterium which induces hairy root disease of dicotyledonous plants, is used to induce hairy roots. In roots transformed with this bacterium, a segment of the bacterial DNA, named the region T (transferred DNA) of the plasmid Ri (root inducing) is incorporated into the host plant cells (Chilton et al. 1982). Integration and expression of this DNA in the plant genome lead to the development of the hairy root phenotype and synthesis of novel low molecular weight compounds called opines (Tepfer and Tempé, 1981). Depending on the strain of *A. rhizogenes* used for the transformation, different principal opines can be found in the tissues of the hairy root: agropine, mannopine, cucumopine or mikimopine (Dessaux et al. 1992). In addition, depending on the gene incorporated in the plant genome, the root can have a change of geotropism. Some hairy roots show a highly negative geotropic nature, some only have a slightly negative geotropic behaviour while others keep their positive geotropism. This is due to a change of auxin sensitivity and the redistribution of this hormone after root transformation (Legué et al. 1996). Roots with a negative geotropism should be incubated in inverted position, to make the roots grow inside the medium.

In this section, only the transformation procedure of carrot (*Daucus carota* L.) with *A. rhizogenes* A4 is described. For the transformation procedures of other plants used as hosts in monoxenic cultures, we refer to the references given in Table 1 of Chapter 7.

5.2.1
Material

Equipment

- Laminar flow hood
- Cooled heated incubator
- Bunsen burner or bead sterilizer

Laboratory material

- Forceps
- Scalpel
- Petri plates filled with 1% water agar
- Petri plates filled with modified White's (Bécard and Fortin 1988) medium supplemented with 500 mg/l carbenicillin
- Small plastic wrap roll

Disinfection solutions

- 95% (v/v) ethanol
- 1% NaOCl
- Distilled water

Bacterial strain

- Agrobaterium rhizogenes strain A_4

5.2.2
Procedure (Following Bécard and Fortin 1988)

1. Thoroughly wash the carrots in water, peel them, dip in 95% (v/v) ethanol for 10 s, and surface sterilize in 1% NaOCl for 15 min.
2. Rinse the carrots in sterile distilled water and cut transversely into 5-mm-thick slices with a sterilized scalpel.
3. Place the slices in Petri plates filled with 1% water agar and inoculate with the A_4 *A. rhizogenes* strain on the distal face of the sections. The bacterial suspension is taken from a 2-day-old culture grown on Difco Nutrient agar.
4. After 3 weeks, some transformed roots proliferate on the inoculated sections. Aseptically excise and transfer the roots to Petri plates containing modified White's medium, supplemented with 500 mg/l carbenicillin. Three successive sub-cultures are necessary to free the transformed roots of bacteria.
5. Finally, excise one root apex from the final subculture (minimum length 3 cm) and place it on fresh modified White's medium to initiate a clonal culture.
6. Seal the Petri plates with plastic wrap and incubate in an inverted position in the dark at 27 °C.
7. Maintain the excised root on MSR medium as described below.

5.3
Host Root Cultivation

5.3.1
Materials

Equipment

- Laminar flow hood

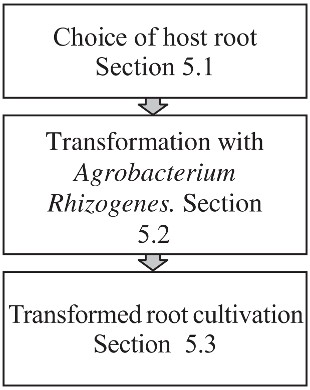


Fig. 3. General scheme of the transformed carrot root organ culture

- Cooled heated incubator
- Bunsen burner or bead sterilizer (cf. Fig. 3)

Laboratory material

- Petri plates filled with MSR medium
- Forceps
- Scalpel
- Small plastic wrap roll

5.3.2
Procedure

1. Every 3–4 weeks, carefully remove white, healthy, turgescent non-ramified apexes (with a minimum length of 3 cm) from the Petri plate.
2. With forceps, place two roots with actively growing apexes in a 9-cm-diameter Petri plate containing fresh MSR medium (the two apexes should be placed 'head to tail' to favour their growth in opposite directions, thereby covering the whole plate) (Fig. 4).
3. When the roots are placed on the medium, push lightly on the apexes to encourage them to intimately contact the gel. Proceed very carefully to avoid breaking the apex.
4. Seal the Petri plate with plastic wrap and incubate in a inverted position at 27 °C in the dark.

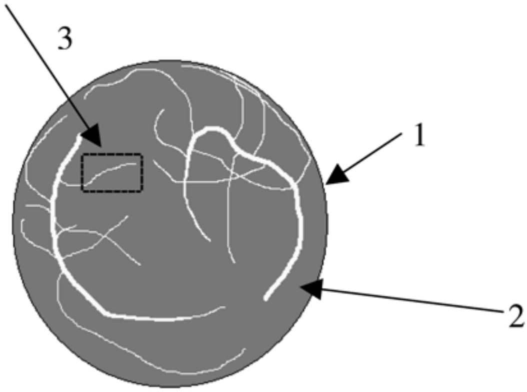


Fig. 4. Representation of a 4-week root organ culture. 1 Main root, 2 secondary root, 3 apex suitable for subcultivation

6
AM Fungal Propagules: Selection and Disinfection

6.1
Selection of Propagule

As described in Fortin et al. (2002), two types of propagule can be used to initiate monoxenic cultures: (1) spores or sporocarps, and (2) mycorrhizal root fragments containing vesicles or isolated vesicles. Depending on the AM fungi, often one type of propagule is better adapted to initiate a monoxenic culture (Table 1). For *Glomus* species producing small spores (diameter $< 100\,\mu$m), disinfection of spores is often difficult without damage. The germination rate and hyphal re-growth are generally low or even absent. Species producing such spores habitually also produce a high number of vesicles within roots, i.e. *G. intraradices*, *G. proliferum*, which therefore should be considered the preferred propagules to initiate the monoxenic culture. Germination of vesicles either isolated or within the root segment is fast and efficient. In the case of mycorrhizal root segments, multiple germination arises from the root extremities (Diop et al. 1994, Declerck et al. 1996), increasing the infective potential.

For species producing few (some Glomeraceae species, *G. caledonium*, *G. mosseae*, etc.) or no vesicles (the Gigasporaceae) and/or big spores (diameter $> 150\,\mu$m), spores should be used. When the fungus produces sporocarps, these are always the most appropriate propagules to initiate the monoxenic culture because of their multiple germination (Fig. 5).

Remark: When monoxenic monosporal cultures are required (in phylogeny studies, for example) from species established in monoxenic culture from mycorrhizal root fragments, it is necessary to subculture single isolated daughter spores with a new host root. Only the first generation initiated with this daughter spore and subsequent generations will be adapted for such studies.

6.2 Disinfection Process

For a successful disinfection of spores and root pieces, a combination of antibiotic treatments should be applied on the extracted material (spores or root pieces). The successful combination of chloramine T and streptomycin was developed for *Glomus* species by Mosse (1962) and later a combination of chloramine T, streptomycin and gentamicin with the inclusion of the surfactant Tween 20 was reported by Mertz et al. (1979) for *Gigaspora* species. As an alternative, or as supplement to the use of antibiotics, sodium hypochlorite was used by Daniels and Menge (1981). Nowadays, sodium hypochlorite is replaced by calcium hypochlorite, since it is an instable product which degrades when remaining attached to the fungal propagules. The sodium hypochlorite solution, however, when not rinsed thoroughly, stays around the disinfected spores or root pieces where it can have a toxic effect. Besides this chemical treatment, Strullu and Romand (1986) adjusted a physical treatment, i.e. a treatment with sonications. With this treatment, even the microparticles which remain attached to the root pieces and spores are removed, assuring a thorough cleaning of the material.

6.2.1 General Considerations

Prepare all the material under laminar flow. The vacuum outlet and filtration system (Fig. 6F) should be assembled, the nylon membrane placed on the filter support, and the apparatus autoclaved at 121 °C for 15 min (1 bar pressure).

The disinfection occurs in two main steps (Fig. 5): a chloramine T treatment (step 1) and an antibiotic treatment (step 2). Prior to the disinfection process and depending on the propagule, i.e. spore or root fragment, and on the complexity of its structures, e.g. ornamented spores which influence the level of contamination, it is necessary to resort to pre-treatment steps (Fig. 5) or to adapt the duration of contact with each solution.

Some species like *Gi. margarita* and *Gi. gigantea* require a cold treatment prior to disinfection. In this case, the spores are placed at 4 °C (3 weeks) prior to isolation and disinfection (Fortin et al. 2002).

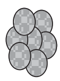
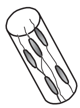
Species type	Inoculum	The desinfection occurs in	Pretreatment 1	Pretreatment 2	Step1 Chloramine T treatment	Step2 Antibiotics treatments
Species producing large spores eg. Gigasporaceae – Glomaceae eg. *Glomus caledonium, mosseae*, etc.	Spore 	Vacuum filtration apparatus	None	None	10 min (2-10min)[1]	10 min (2-10 min)[1]
Species producing sporocarps : Glomaceae : *Glomus mosseae*	Sporocarp 	Vacuum filtration apparatus	Ultrasonic 1 min	None	10 min (2-10min)[1]	10 min (2-10min)[1]
Species with high root colonization and Vesicles production : Acaulosporacea-Glomaceae eg. Gl*omus intraradices, fasciculatum, proliferum*, etc.	Root fragments 	(Ultrasonic bath (half of the time))[2]	(Ethanol 98 % 10s)[2]	Ca-hypochlorite 6% 2 min	10 min (2-10min)[1]	10 min (2-10min)[1]

Fig. 5. Summary of the disinfection procedure. *1* Exposure time must be adapted to the contamination level of the starting material and the sensibility of the structures. *2* These steps can be suppressed if the root is not too contaminated and fragile

6.2.2 Material

Equipment

– Laminar flow hood

- Vacuum filtration apparatus (Whatman®)
- Sonicator
- Stereo microscope
- Cooled heated incubator
- Bunsen burner or bead sterilizer

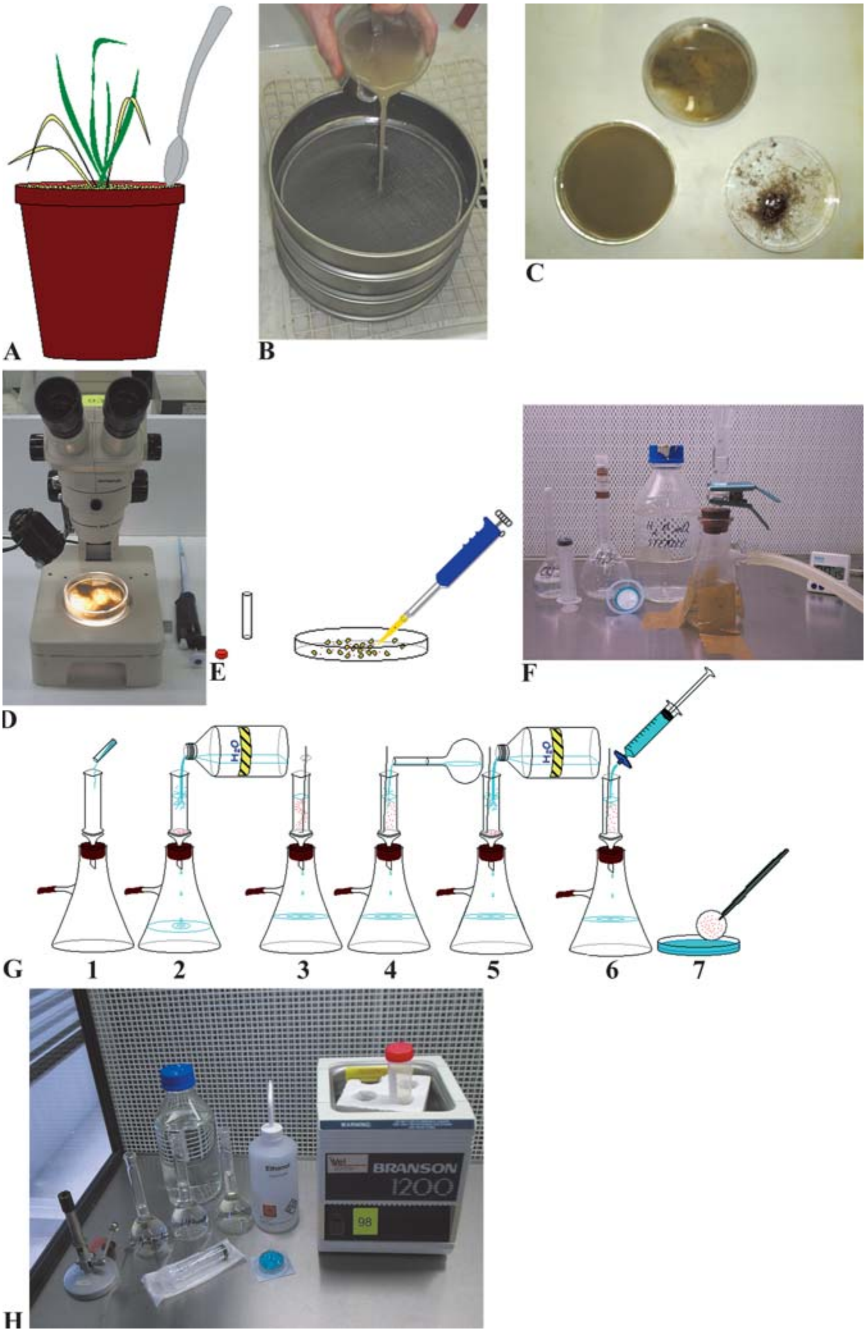
Laboratory material

- Sterile Petri plates (90 mm diameter)
- Petri plate filled with MSR medium
- 20-ml sterile syringe
- 0.2-µm sterile filter (Acrodisc GelmanSciences)
- Sterile nylon membrane with 0.45-µm pores, 25-mm diameter (adapted to the vacuum apparatus; GelmanSciences)
- Sterile Pasteur pipette
- Spoon
- Forceps
- Scalpel
- Needles
- Adjustable volume pipette and sterile tips
- Small plastic wrap roll
- 50-ml sterile centrifuge tube (falcon type) or sterilized small beaker
- 50-ml floating (polystyrene) support for centrifuge tube (or small beaker)
- Ground clay (Terragreen®)

6.2.3 Solutions for Disinfection

For about 100 spores, 30 sporocarps or 10 root fragments:

1. Sterile water 1 l.


A
B
C
D
E
F
G
1
2
3
4
5
6
7
H
BRANSON
1200
98

◀ **Fig. 6A–H.** Illustration of the disinfecting process. **A–G** refer to spores/sporocarps (see Sect. 6.2.4), and **H** refers to root fragments (see Sect. 6.2.5). **A** Sampling of spores from a pot culture. **B** Sieving of soil sample. **C** Soil fractions after sieving. **D**, **E** Selection of spores for the disinfection process. **F** Illustration of the disinfection material (spores/sporocarps). **G** Scheme of the disinfecting process for spores (sporocarps). 1 Transfer of the propagules to the filtration apparatus, 2–5 rinsing with water, 3 agitation, 4 chloramine T treatment (step 1), 6 antibiotic treatment (step 2), 7 transfer to antibiotics solution. **H** Illustration of the disinfection material (root fragments)

2. Ethanol 95–98% (100 ml) for pre-treatment 1 and disinfection of the material.
3. Calcium hypochlorite solution 2% (= 20 g/l, 100 ml) for pre-treatment 2.
   – *Remark*: When the calcium hypochlorite is not totally dissolved, the solution must be filtered through a Whatman® filter paper no. 1.
4. Chloramine T 2% solution (= 20 g/l) with two drops of Tween 20 (should be added before use; 100 ml; solution for step 1).
5. Antibiotics solution: Streptomycin sulphate 0.02% (20 mg/100 ml) and gentamicin sulphate 0.01% (10 mg/100 ml; 100 ml; solution for step 2).
   – *Remark*: This must be filtered with a 0.2-μm acrodisc filter (GelmanSciences) before use.

6.2.4 Spores, Sporocarps

Sampling of Inoculum.
Spores or sporocarps can be collected from a field soil sample or from a pot culture.

1. Sample approximately 100 ml of soil (growth substrate) with a spoon from a pot culture and transfer to a plastic container or a small beaker. The soil sample should be homogeneous, thus recovered from the complete volume of the pot culture. Refill the hole with fresh growth substrate (Terragreen®; Fig. 6A).
2. Transfer the soil to a 1-l beaker and add 500 ml of demineralized water. Mix gently and before complete settling transfer the supernatant to the top of a sequence of sieves with cutoffs 500, 106 and 38 μm (Fig. 6B). This combination is suitable to collect AM fungal spores; the upper fraction will retain soil and root debris but also mycorrhizal roots, while both other fractions will retain spores belonging to all the genera.

3. Transfer each fraction to a glass Petri plate with a small amount of water (Fig. 6C). Pick up the spores ($\sim$ 100) or sporocarps ($\sim$ 30) with an automatic pipette under a binocular microscope (Fig. 6D). Wash the spores several times in water until complete elimination of fine debris is achieved. The debris-free spores (or sporocarps) are transferred to a small tube containing water (Fig. 6E).

Remark: When few spores are present in the sample, it may be more convenient to use the centrifugation method based on the establishment of a gradient using a highly concentrated substance like sucrose, glycerol or Percoll etc. (Mertz et al. 1979; Furlan et al. 1980; Hosny et al. 1996).

Disinfection Procedure.

1. To eliminate soil debris from the peridium of the sporocarps, treat them in an ultrasonic bath for 1 min. Usually, such pre-treatment is not required for spores, unless they are ornamented.
2. Transfer the cleaned spores or sporocarps to the filtration apparatus connected to a vacuum outlet (Fig. 6F and G, image 1), rinse them twice with sterile water (Fig. 6G, image 2) and agitate gently with a Pasteur pipette (Fig. 6G, image 3). Water outflow should be constant and slowed down by the end of the process to eliminate the liquid, but avoiding complete drying of the membrane.
3. Treat the spores or sporocarps with chloramine T 2% solution (with 2–3 drops of Tween 20), and filter it very slowly through the spores or sporocarps for a 10-min period. Continue manual agitation with the Pasteur pipette during the treatment (step 1, Fig. 6G, image 4).
4. Wash the spores three times with sterile water (Fig. 6G, image 5).
5. Treat the spores with the antibiotic solution for 10 min by following the same process as described above (step 2, Fig. 6G, image 6).
6. When the solution is removed, interrupt the filtration carefully and transfer the membrane supporting the disinfected spores or sporocarps into a plastic Petri plate containing the filtered antibiotics solution (Fig. 6G, image 7).
   Remark: After the solution is removed, the membrane must be wet but without liquid film on its surface which can cause the loss of spores when the apparatus is disassembled.
7. Under a binocular microscope, select healthy-looking spores and place them in Petri plates containing the MSR medium (five spores or one sporocarp per plate; enough space should be left between the spores to facilitate their transfer after germination; Fig. 7A).

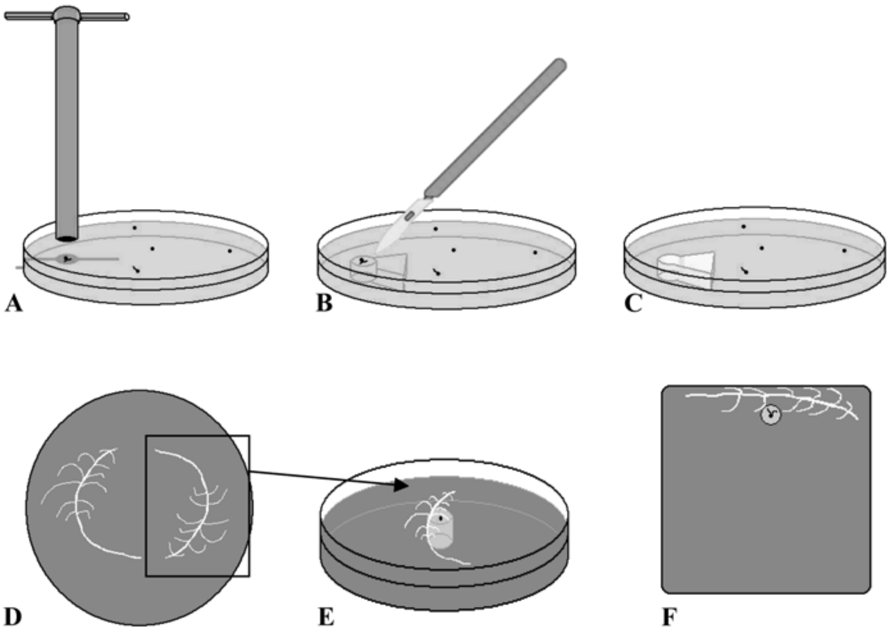


Fig. 7 A–F. Establishment of a monoxenic culture from a disinfected spore. **A–C** Extraction of a disc of gel supporting the germinated spore. **D** Selection of a suitable host root. **E** Placement of the propagule and the host root in a round Petri plate. **F** Placement of the propagule and the host root into a square Petri plate

8. Seal the Petri plates with plastic wrap and incubate at 27 °C in the dark.

6.2.5
Root Fragments, Isolated Vesicles

Sampling of Inoculum.
The best results are obtained with young, healthy roots containing numerous vesicles. This type of material is not easy to collect under field conditions, and mycorrhizal roots from trap plants cultured in the greenhouse are more suitable to initiate a monoxenic culture. Leek plants are a convenient host because the roots are white and translucent, making the vesicles of the fungus more easily observable.

1. Sample roots from pot culture with a forceps.
2. Under a stereo-microscope, select between 10 and 20 root fragments (∼ 5 cm length) which containing numerous vesicles.

3. Rinse them with demineralized water and store in a sterile 50-ml tube (or in a sterile glass beaker) in sterile water (under laminar flow).

Disinfection Procedure.
The sterile tube is placed in an ultrasonic bath on a floating support. The apparatus is alternatively switched on half of the time for each disinfection step (Fig. 6H).

1. Clean the roots thoroughly with sterile (demineralized) water, three times.
2. Treat the roots with alcohol (95–98%) for 10 s (pre-treatment 1) (Fig. 5).
3. Rinse three times with sterile (demineralized) water.
4. Treat the roots with the calcium hypochlorite solution for 1–2 min (pre-treatment 2) (Fig. 5).
5. Rinse three times with sterile (demineralized) water.
6. Treat the roots with the solution of chloramine T 2% (20 g/l, with 2–3 drops of Tween® 20) for 10 min (step 1).
7. Rinse 3 times with sterile demineralized water.
8. Treat the roots with the antibiotic solutionsm for 10 min (step 2) (Fig. 5).
9. Remove the antibiotic solutions and transfer the roots to a Petri plate with a fresh antibiotic solution. Keep them in the antibiotic solution until use.
10. Place one root fragment of 1-cm length, containing numerous vesicles (selected under the binocular microscope), onto a Petri plate filled with MSR medium.
11. Seal the Petri plates with plastic wrap and incubate at 27 °C in the dark.

Enzymatic Treatment for Vesicle Isolation.
To obtain isolated vesicles, the incubated root fragments, disinfected following the method described above, are treated with an enzymatic solution. This solution degrades the root and, as a result, the mycorrhizal structures, i.e. vesicles, intraradical hyphae and arbuscules, are released. Vesicles can be selected under a binocular microscope and cultivated in vitro on synthetic medium.

Procedure (following Strullu and Romand 1987).

1. Prepare the enzymatic solution: macerozyme R10: 0.2 g/l, cellulase R10 : 1 g/l, driselase: 0.5 g/l in 1 l of liquid culture medium.
   – *Remark*: This must be filtered with a 0.2-μm acrodisc filter (GelmanSciences) before use.
2. Pour 5 ml of this solution into a Petri plate (45-mm diameter) and add one drop of streptomycin (5 ml/l).
3. Place about 10 root fragments of 1-cm length into each Petri plate.
4. After 24 h, the root fragments are transferred to Petri plates containing the same liquid culture medium without enzymes.

Different protocols for the liberation of mycorrhizal structures in root fragments are fully described in Saito (1995) and Solaiman and Saito (1997).

Remark: Isolated vesicles can also be obtained using a physical treatment. The disinfected root fragments are in this case pierced with a needle. When the root fragment is opened, the vesicles are released into the surrounding solution.

7 Monoxenic Culture Establishment

7.1 Germination of Disinfected Propagules

The germination of spores and sporocarps often occurs in a period varying between 2 and 30 days after disinfection. In general, spores and sporocarps do not require any specific conditions for germination. However, it has been observed for some AM fungal strains that germination was stimulated by the presence of a root or CO_2 (Bécard and Piché 1989). Therefore, for some strains, the propagules are placed in the vicinity of a root. It was also demonstrated for some *Gigaspora* species (e.g. *Gi. margarita* and *Gi. rosea*) that cold treatment is always required before isolation and disinfection (see Sect. 6.2.1).

For mycorrhizal root fragments, the hyphal re-growth from the extremities of mycorrhizal root segments is usually observed within 2–15 days. Germination of vesicles occurs 2–10 days after disinfection.

For both inoculum types, spores/sporocarps and mycorrhizal root segments or isolated vesicles, it is important to associate the germinated propagules with a host root shortly after germination to avoid damage of the newly formed hyphae during the association process and to increase colonization aptitude.

7.2 Material

Equipment

- Laminar flow hood
- Stereo-microscope
- Cooled heated incubator
- Bunsen burner or bead sterilizer

Laboratory material

- Sterile Petri plates
- Petri plate filled with MSR medium
- Forceps
- Scalpel
- Needles
- Cork borer (0.05 to 0.25 cm diameter)
- Adjustable volume pipette and sterile tips
- Small plastic wrap roll

7.3 Association Methods

1. Select a host root (transformed carrot root) of 8-cm length with few apexes forming a herringbone structure from a 2–3 week old root organ culture (see Fig. 7D).
2. If the inoculum is a spore (since five spores were placed in the same plate for germination):
   a) From a Petri plate filled with the MSR medium, remove a disc of agar with a cork borer. This disc should be identical in size to the disc of gel supporting the germinated spore.
   For *Glomus* and *Scutellospora*, the disc of gel should be removed from the centre of a mono-compartmental Petri plate (9-cm diameter) or from the centre of the proximal compartment of a bi-compartmental system (for *Glomus*).
   For *Gigaspora* species, the disc of gel should be removed from the centre of the upper part of a square mono-compartmental plate. The diameter of the disc should be large enough to include the spore and germinating hyphae (9-mm diameter is usually enough; Fig. 7A–C).

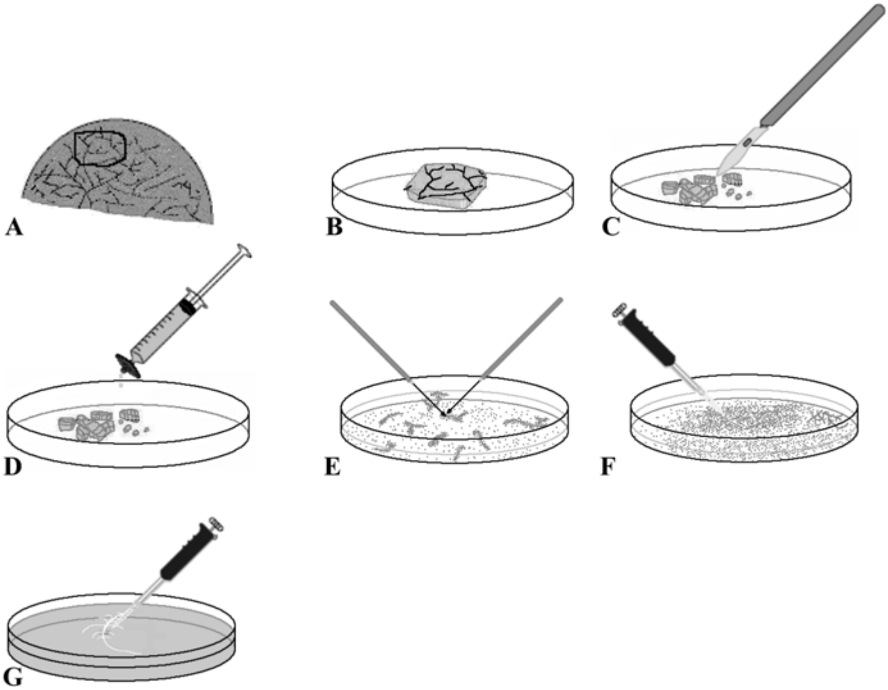


Fig. 8A–G. Illustration of subcultures from dissolved spores. **A** Selection of a piece of gel containing spores. **B–D** Addition of 10 mM citrate buffer to dissolve the gel. Spores can be better separated when the gel piece is fragmented. **E** Separation of spores in small clusters. **F**, **G** Host root inoculation

b) Transfer the disc of gel supporting the germinated spore of *Glomus* and *Scutellospora* (Fig. 7D, E) or *Gigaspora* (Fig. 7F) into the hole of the removed disc.

c. Continue with step 3.

3. Place the root carefully in the near vicinity of the propagule (spore, sporocarp or mycorrhized root fragment) into the Petri plate, so that the growing hyphae are perpendicular to a secondary root (see Fig. 7E, F).
4. Seal the Petri plates with plastic wrap and incubate in an inverted position at 27 °C in the dark for the *Glomus* and *Scutellospora* species in mono- or bi-compartmental systems. The *Gigaspora* cultivated in a square Petri plate has to be incubated vertically (root in the upper part) at 27 °C in the dark.

8 Continuous Culture

8.1 Association Establishment

For a *Glomus* species like *G. intraradices*, a mycelium appears after a few days and new spores are produced very quickly thereafter. The mycelium grows extensively, rapidly invading the complete volume of the Petri plate. In the bi-compartmental system, the sporulation is higher in the distal compartment containing no carbon (see Sect. 3). The sporulation capacity of the fungal strain and the development of mycelium depend on the strain. After 3 months, a culture of *G. intraradices* can produce more than 10,000 spores.

For other *Glomus* strains, for example, *G. caledonium*, which produces large spores and few vesicles, mycelium can grow rapidly and extensively but spores appear very slowly (after one or more months).

For Gigasporaceae (*Gi. rosea*, *S. reticulata*, etc.), the production of mycelium is less profuse, auxiliary cells appear rapidly, but it takes several weeks for spores to be produced (Fortin et al. 2002).

8.2 Continuous Culture

The first continuous culture (see definition in Chap. 1) of *Glomus* species was achieved by Strullu and Romand in 1986 using the intraradical form, and was thereafter extended to various AM fungi (Table 1). The continuous culture is obtained by associating monoxenic mycorrhizal roots and/or spores (Chabot et al. 1992), often attached to extraradical mycelia, to a new, actively growing non-mycorrhizal host root, onto a fresh medium. This first method is largely used and is effective for a range of *Glomus* species (Diop et al. 1994a; Strullu et al. 1997; Declerck et al. 1998, 2000) and for *Scutellospora reticulata* (de Souza and Declerck 2003; Table 1).

A second method has been used by St-Arnaud et al. (1996) and is effective for *Glomus* species having a well-developed intraradical phase, such as *G. intraradices*. In this method, apical segments of actively growing mycorrhizal roots with or without extraradical mycelium-supporting spores are transferred to a fresh medium. The root and associated fungus continue to proliferate across successive transfers onto fresh medium. This procedure requires the use of young, actively growing cultures, to allow the continuous growth of the host root.

For *Glomus* species producing few spores and vesicles, continuous cultures are difficult to obtain. For *Gigaspora* species they were only reported as unpublished results in Fortin et al. (2002). Hence, for this latter species, it is necessary to periodically start new monoxenic cultures with disinfected spores from pot cultures.

8.3 Material

Equipment

- Laminar flow hood
- Stereo-microscope
- Cooled heated incubator
- Bunsen burner or bead sterilizer
- Agitator (preferably rotating agitator)

Laboratory material

- Sterile Petri plates
- Petri plate filled with MSR
- 20-ml sterile syringe
- 0.2-µm sterile filter (Acrodisc GelmanSciences)
- 50-ml sterile centrifuge tube (falcon type)
- Forceps
- Scalpel
- Needles
- Adjustable volume pipette and sterile tips
- Small plastic wrap roll

8.4 Solutions

1. Citric acid stock solution: 0.1 M (autoclaved).
2. Sodium citrate stock solution: 0.1 M (autoclaved).

3. Citrate buffer (0.01 M): under laminar flow, mix 0.018 volume citric acid solution, 0.082 volume sodium citrate solution, complete the volume with sterile water, and adjust the pH to 6 if necessary with NaOH 1 M. The solution is filtered through an acrodisc 0.2 mm before use.
4. Sterile water.
5. Ethanol for material disinfection.

8.5 First Method: Propagule Re-Association

8.5.1 Isolation of Propagules

Propagules can be extracted from the gel in several ways:

1. If spores are relatively big (> 150 µm) and occur in small quantity (such as *Gigaspora*, *Scutellospora* and some *Glomus* species), it is easy to extract a single spore attached to the extraradical mycelium with a needle. Only one spore is necessary to start a new culture but, to increase the chances of association, around five spores can be use as inoculum.
2. If spores are numerous and of small to medium size, such as reported for the majority of the *Glomus* species cultured monoxenically, simply cut a small piece of gel containing large amount of spores (i.e. 1–5 cubes of 0.5 cm^3, $\sim$ 200 spores) attached to the extraradical mycelium.
3. Alternatively, for species producing high numbers of spores, the gelling agent may be removed from the culture medium, to stimulate re-growth of the fungus. The gelling agent (=agar substitutes like Gelgro or Phytagel but not agar) can be dissolved by a citrate buffer (10 mM; Fig. 8) as follows:
   a. Extract a piece of gel containing approximately 200 spores from the mother culture and transfer it to an empty sterile Petri plate (diameter 9 cm; Fig. 8A–C).
   b. Add 10× the volume of citrate buffer filtered through a 0.2-µm acrodisc (for example, for a gel block of 0.5 g, add 5 ml of buffer) and seal the plate with plastic wrap.
   Remark: Do not put too much liquid in the Petri plate, to avoid any risk of contamination (for example, in a Petri plate of 9-cm diameter, put a maximum of 25 ml citrate buffer; Fig. 8C, D).

3. Agitate the Petri plate slowly on a rotating agitator (50 rotation/min) at 25–27 °C for 30 min to 1 h, depending on the volume of gel to dissolve.
4. Transfer the spores attached to the extraradical mycelium to a new Petri plate containing sterile water using a binocular microscope (under laminar flow), then separate the spores into individuals or little clusters attached to the extraradical mycelium (containing 15–30 spores, depending on spore size; Fig. 8E).

8.5.2
Association with a Root Host

1. Place a transformed carrot root suitable for association (8-cm length, with about six secondary roots of 1–2 cm length) in a Petri plate filled with the MSR medium. In the mono-compartmental culture system using square Petri plates (suitable for *Gigaspora* species), the root is placed in the upper part of the Petri plate; in the other models (round mono- and bi-compartmental), the root is placed in the centre of the root compartment.
2. Place the medium pieces containing spores attached to the extraradical mycelium, or the single spore (or spores clusters) and/or mycorrhizal root pieces homogeneously along the root, in the vicinity of secondary roots, to favour the association (Fig. 8F, G). Association generally occurs near the growing apex where the cellular wall is thinner and easier to penetrate.
3. Seal the Petri plate with plastic wrap and incubate at 27 °C in the dark in an inverted position (vertically for *Gigaspora* species).

Germination and association usually occur between the 3rd and the 15th day after association.

8.6
Second Method: Mycorrhized-Apex Transfer

For *Glomus* species showing a well-developed intraradical phase, the easiest way to obtain a continuous culture is to transfer a section (2×2 cm) of medium containing healthy root apexes with extraradical hyphae and spores onto a Petri plate containing MSR medium. The mycorrhizal root continues to grow and the fungus proliferates.

Remark: It is necessary to use a young culture (some weeks). When the culture is too old, the medium becomes exhausted and the root becomes necrotic.

9 Conclusion

The protocols presented here aim to give researchers the possibility to initiate and maintain their own AM fungal monoxenic cultures. They should encourage general use of such fungal materials, enlarging the diversity of species cultured in this system. The undisputable benefits derived from this culture system are highlighted in all chapters of this book. This considerable impact is not only restricted to the study of symbiotic interactions, but also permits the increase in knowledge in the morphology, taxonomy, phylogeny and biochemistry fields together with some aspects of their ecology.

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